1. **Write a Python program to draw line charts of the financial data of Alphabet Inc. between October 3, 2016 to October 7, 2016.**
   Sample Financial data (fdata.csv):
   Date,Open,High,Low,Close
   10-03-16,774.25,776.065002,769.5,772.559998
   10-04-16,776.030029,778.710022,772.890015,776.429993
   10-05-16,779.309998,782.070007,775.650024,776.469971
   10-06-16,779,780.47998,775.539978,776.859985
   10-07-16,779.659973,779.659973,770.75,775.080017
2. **Write a Python program to Program to show number of students vs marks obtained (Horizontal Bar Plot)**
   nos = [2, 9, 20, 25, 30, 39]
   marks = [12, 24, 25, 27,29, 30]

3. **#Example 6: The file "monthsales.csv" have stored the sales (in Rs) made in first six months for four years.**

2018	2019	2020	2021
2500	2600	2450	3250
2200	1800	2000	3200
3000	2500	3000	3800
2800	3000	3100	2800
2750	3200	1950	2900
3500	2000	3550	2300

Year-wise sales of first six months

Draw the line chart for the data given above with following details.

1. **Chart title as "Year-Wise Sales"**
2. **X-axis label as "Months"**
3. **Y-axis label as "Sales"**

4. **Assuming the same CSV file, i.e., monthsales.csv, plot the line chart with following customisations. Chart should have Month name on X-axis instead of numbers.**
   Maker ="*"
   Marker size=10
   linestyle="–"
   Linewidth =3
5.

Anant is working as a Data Analyst in a company named “”Compu-Tech”. He stored the following data in a file named “emp.csv”. He wants to present “Department” (First Column) and “No_of_Emp” (Third Column) in a bar chart. Help him to write the code.

Department	**Expenditure (in Rs)**	**No_of_Emp**
HR	**2500**	**40**
Sales	**3000**	**30**
Finance	**2200**	**35**
Marketing	**5100**	**32**
IT	**3900**	**45**

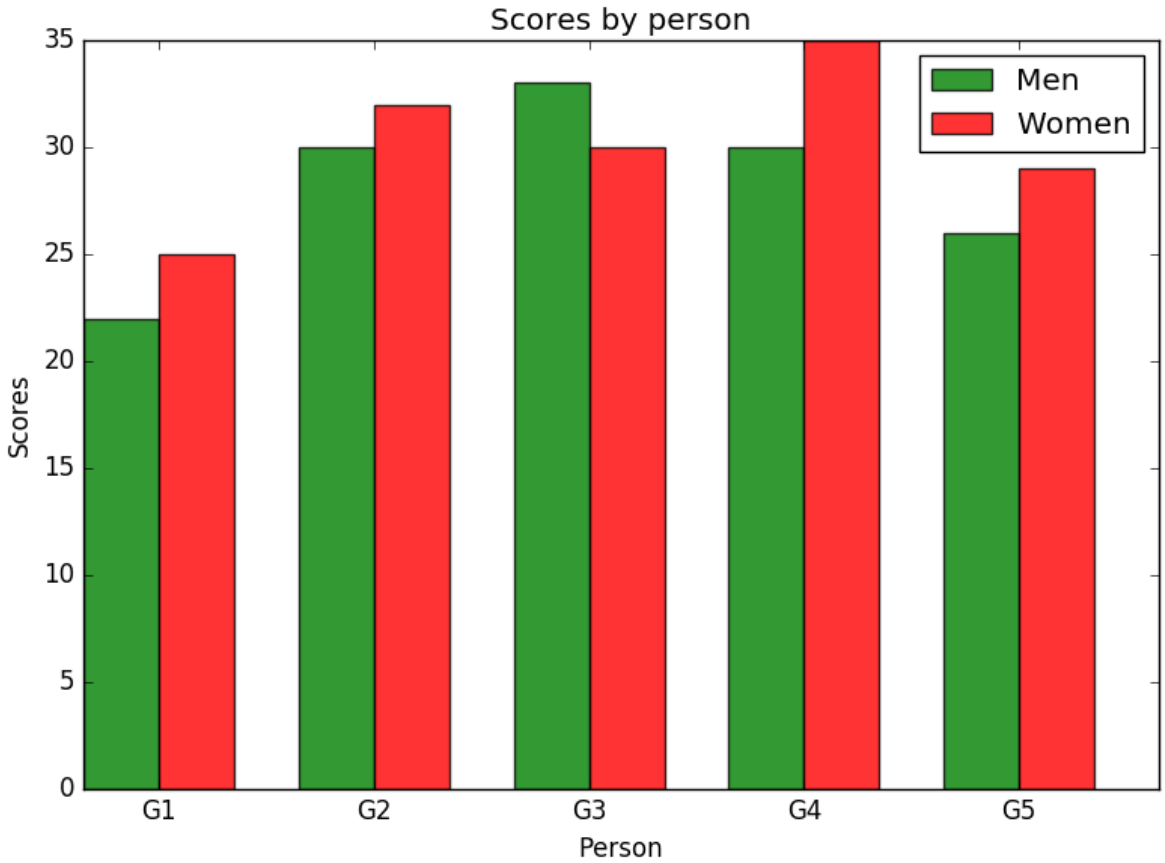
- **Chart Title: Compu-Tech Employee Data**
- **X-Axis Label: Departments**
- **Y-Axis Label: Number of Employees**
- **Department should be displayed on X-axis**
- **Colour of bar should be RED**

6. Visit data.gov.in , search for the following in “catalogs” option of the website:
   • Final population Totals, India and states
   • State Wise literacy rate Download them and create a CSV file containing population data and literacy rate of the respective state.
   Also add a column Region to the CSV file that should contain the values East, West, North and South.
   Plot a scatter plot for each region where X axis should be population and Y axis should be Literacy rate. Change the marker to a diamond and size as the square root of the literacy rate. Group the data on the column region and display a bar chart depicting average literacy rate for each region.
7. In order to assess the performance of students of a class in the annual examination, the class teacher stored marks of the students in all the 5 subjects in a CSV “Marks.csv” file as shown in Table. Plot the data using boxplot and perform a comparative analysis of performance in each subject.

Name	English	Maths	Hindi	Science	Social_Studies
Rishika Batra	95	95	90	94	95
Waseem Ali	95	76	79	77	89
Kulpreet Singh	78	81	75	76	88
Annie Mathews	88	63	67	77	80

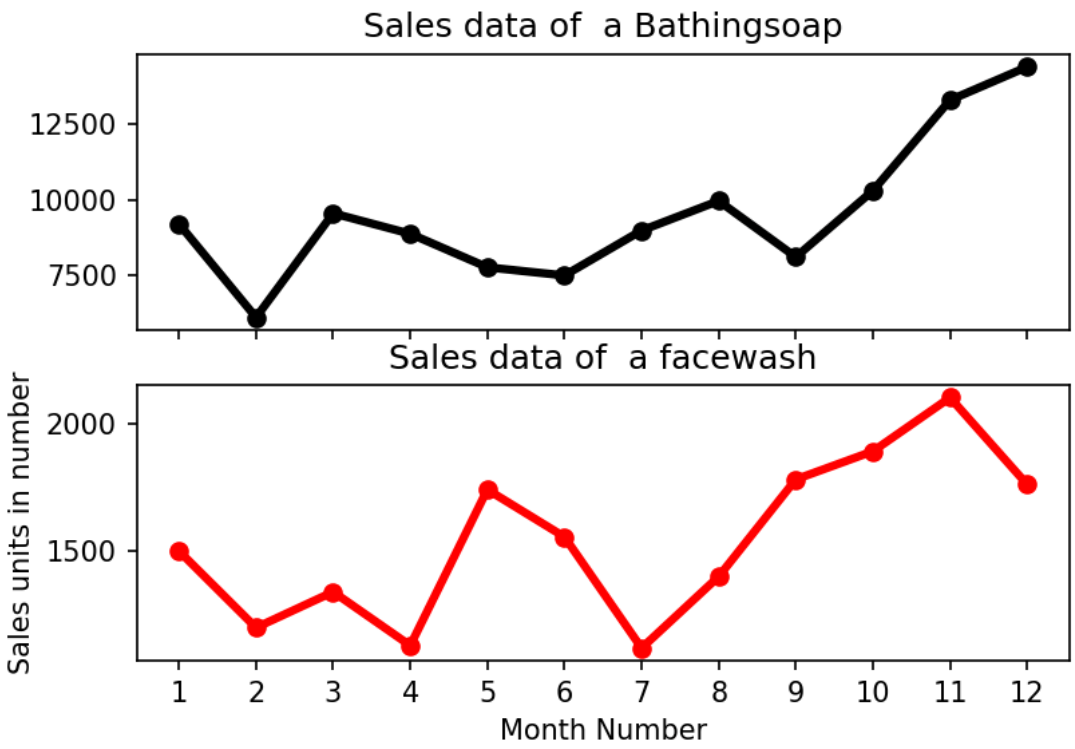
8. Write a Python program to create bar plot of scores by group and gender. Use multiple X values on the same chart for men and women.

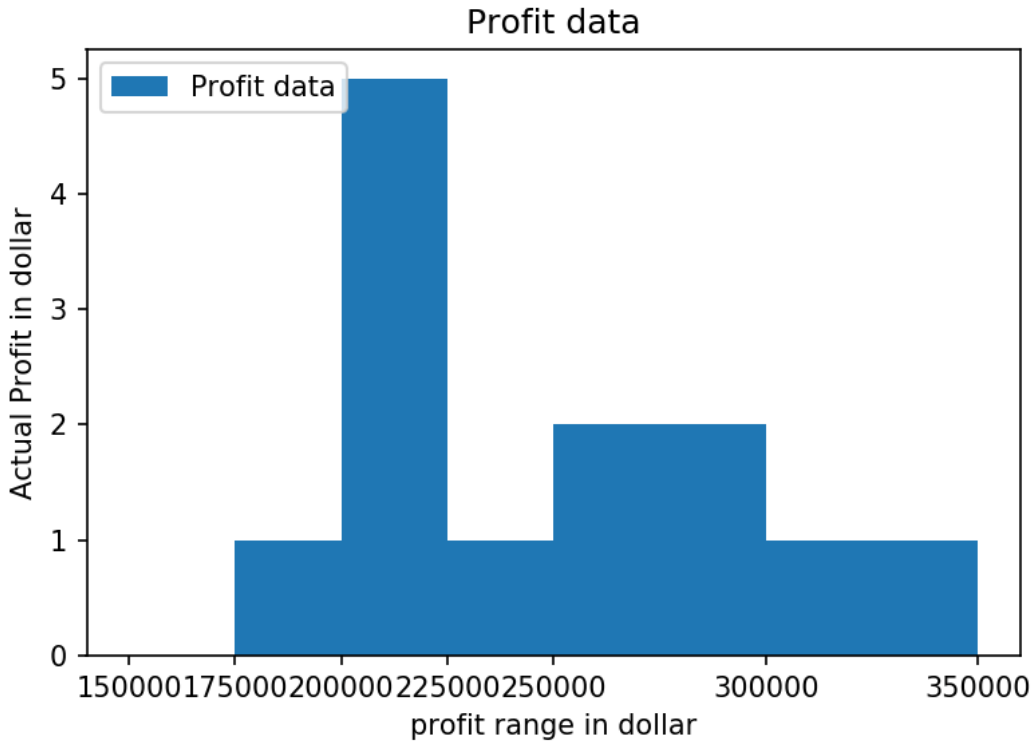
   Sample Data:
   Means (men) = (22, 30, 35, 35, 26)
   Means (women) = (25, 32, 30, 35, 29)
   The code snippet gives the output shown in the following screenshot:



9. Use the following CSV file for this exercise
   https://drive.google.com/file/d/1FTV7KLn_PzOjVHrT9--5hMaPKDfmr13X/view?usp=sharing
   a. Calculate total sale data for last year for each product and show it using a Pie chart
      Note: In pie chart display Number of units sold per year for each product in percentage

b. Read Bathing soap facewash of all months and display it using the Subplot The Subplot should look like this.



10.Read the total profit of each month and show it using the histogram to see the most common profit ranges

The histogram should look like this.